

AI HOSPITAL RECEPTIONIST

Virtual Triage & Registration System

1. Background & Context

Healthcare facilities in high-volume and remote areas are frequently crippled by registration bottlenecks. A chronic shortage of frontline staff forces patients to wait extended periods before their initial consultation, often while in significant distress. This inefficiency leads to patient frustration and, more critically, the risk of misrouting urgent medical cases.

The **AI Hospital Receptionist** leverages natural language processing to act as a 24/7 autonomous point of entry. By intelligently classifying patient needs and automating data collection, it ensures that triage is both accurate and immediate, operating without the need for constant human supervision.

2. Project Objectives

- **Universal Accessibility:** Deploy a natural language interface (text/speech) available across kiosks, tablets, and web browsers.
- **Automated Triage:** Utilize AI to classify patients into three key streams: **General**, **Emergency**, or **Mental Health**.
- **Precision Data Collection:** Implement a polite, step-by-step conversational flow to gather demographics and symptoms.
- **Seamless Integration:** Trigger real-time notifications via webhooks and secure cloud storage to alert medical staff instantly.

3. Target Beneficiaries

- **Patients:** Experience reduced wait times and a streamlined, hassle-free registration process.
- **Medical Staff:** Nurses and receptionists are relieved of administrative burdens, allowing focus on clinical care.
- **Healthcare Administrators:** Gain a scalable, low-cost digital triage solution for rural and urban centers alike.

4. Implementation Framework

Phase	Activity Focus	Timeline
Phase 1	Infrastructure Setup: React Frontend, FastAPI Backend, and Supabase integration.	Week 1-2
Phase 2	AI Development: LangGraph workflow, ward classifier logic, and UI polishing.	Week 3-4
Phase 3	Integration: Webhook connectivity, end-to-end testing, and security policy audit.	Week 5
Phase 4	Finalization: Documentation, demo recording, and stakeholder presentation.	Week 6

5. Expected Strategic Outcomes

- **Efficiency:** Registration time is projected to decrease by over **60%**.
- **Reliability:** Guaranteed 24/7 availability eliminates gaps in service during night shifts or staff shortages.
- **Accuracy:** AI-driven classification minimizes human error in ward routing.
- **Scalability:** A flexible cloud-based architecture allows for rapid deployment across multiple hospital chains.

6. Budget Overview

The project is designed with high cost-efficiency, utilizing industry-standard free tiers for the pilot phase:

- **Core Infrastructure (Supabase, Hosting, Webhooks):** \$0 (Free Tier)
- **Domain/Operations (Optional):** ~\$10.00
- **Total Estimated Budget:** **\$0 - \$10.00**

7. Conclusion

The AI Hospital Receptionist is more than a chatbot; it is a full-stack response to a systemic healthcare crisis. By merging modern web technology with LangGraph-based intelligence, this project offers a viable, immediate path toward smarter, faster, and more accessible healthcare.

Contact Information

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